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FACULTAD DE CIENCIAS EXACTAS Y NATURALES
DEPARTAMENTO DE COMPUTACIÓN

Correlatos neurales de los modelos computacionales de búsqueda visual híbrida

Tesis de Licenciatura en Ciencias de Datos

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Buenos Aires, 2026

CORRELATOS NEURALES DE MODELOS DE BÚSQUEDA VISUAL HÍBRIDA

La princesa Leia, líder del movimiento rebelde que desea reinstaurar la República en la galaxia en los tiempos ominosos del Imperio, es capturada por las malévolas Fuerzas Imperiales, capitaneadas por el implacable Darth Vader. El intrépido Luke Skywalker, ayudado por Han Solo, capitán de la nave espacial “El Halcón Milenario”, y los androides, R2D2 y C3PO, serán los encargados de luchar contra el enemigo y rescatar a la princesa para volver a instaurar la justicia en el seno de la Galaxia (aprox. 200 palabras).

Palabras claves: Guerra, Rebelión, Wookie, Jedi, Fuerza, Imperio (no menos de 5).

NEURAL CORRELATES OF COMPUTATIONAL MODELS OF HYBRID VISUAL SEARCH

In a galaxy far, far away, a psychopathic emperor and his most trusted servant – a former Jedi Knight known as Darth Vader – are ruling a universe with fear. They have built a horrifying weapon known as the Death Star, a giant battle station capable of annihilating a world in less than a second. When the Death Star’s master plans are captured by the fledgling Rebel Alliance, Vader starts a pursuit of the ship carrying them. A young dissident Senator, Leia Organa, is aboard the ship & puts the plans into a maintenance robot named R2-D2. Although she is captured, the Death Star plans cannot be found, as R2 & his companion, a tall robot named C-3PO, have escaped to the desert world of Tatooine below. Through a series of mishaps, the robots end up in the hands of a farm boy named Luke Skywalker, who lives with his Uncle Owen & Aunt Beru. Owen & Beru are viciously murdered by the Empire’s stormtroopers who are trying to recover the plans, and Luke & the robots meet with former Jedi Knight Obi-Wan Kenobi to try to return the plans to Leia Organa’s home, Alderaan. After contracting a pilot named Han Solo & his Wookiee companion Chewbacca, they escape an Imperial blockade. But when they reach Alderaan’s coordinates, they find it destroyed - by the Death Star. They soon find themselves caught in a tractor beam & pulled into the Death Star. Although they rescue Leia Organa from the Death Star after a series of narrow escapes, Kenobi becomes one with the Force after being killed by his former pupil - Darth Vader. They reach the Alliance’s base on Yavin’s fourth moon, but the Imperials are in hot pursuit with the Death Star, and plan to annihilate the Rebel base. The Rebels must quickly find a way to eliminate the Death Star before it destroys them as it did Alderaan (aprox. 200 palabras).

Keywords: War, Rebellion, Wookie, Jedi, The Force, Empire (no menos de 5).

AGRADECIMIENTOS

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A mi persona favorita.

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1. INTRODUCCIÓN

- 1.1. Búsqueda visual e híbrida
- 1.2. El modelo nnELM
- 1.3. EEG en búsqueda visual
- 1.4. Pregunta principal y objetivos

2. MATERIALES Y MÉTODOS

- 2.1. Dataset HSEM
- 2.2. Pipeline de análisis de EEG
- 2.3. El modelo nnELM
- 2.4. Extracción de features del nnELM
- 2.5. Asignación de features a fijaciones
- 2.6. Modelo de regresión EEG
- 2.7. Metodología de evaluación

3. ANÁLISIS DE FEATURES

3.1. Validación y análisis de las features

4. RESULTADOS

- 4.1. Resultados con features agregadas
- 4.2. Interpretación de resultados

5. DISCUSIÓN

6. CONCLUSIONES

